Quiz 2 Practice Problem Set: Solutions

MA4.101 Real Analysis, Monsoon 2026

Difficulty. ⋆ warm-up, ⋆⋆ core, ⋆⋆⋆ stretch.

The bracket after each title is the lecture the problem is based on.

How to read the solutions. Boxes marked **KEY METHOD** (orange) are techniques taken from the proofs done in the lectures: they are not asked directly, but variations of them keep appearing in quizzes, assignments and exams, so make sure you can reproduce each one. Boxes marked **KEY PROOF TECHNIQUE** (blue) are general proof patterns such as proving a set equality by two inclusions or an if-and-only-if in both directions. An index of all methods is at the end.

Part A: Real numbers, order and suprema (Lectures 12 to 14)

Problem 1. *Finitely many terms do not matter* [L12] ⋆

Let $a_n = \frac{1}{2} + \frac{(-1)^n}{n}$ for $n \geq 1$, so $a_1 = -\frac{1}{2}$, $a_2 = 1$, $a_3 = \frac{1}{6}, \ldots$

(a) Show (a_n) is a Cauchy sequence of rationals, and show $\mathrm{LIM}_{n\to\infty}\, a_n = \frac{1}{2}$.

(b) The sequence (a_n) is *not* positively bounded away from zero (since $a_1 < 0$). Construct a sequence (b_n) that is positively bounded away from zero and equivalent to (a_n), and conclude that $\mathrm{LIM}\, a_n$ is positive.

(c) State the general principle you used in (b).

(d) Let (c_n) be a Cauchy sequence of rationals with $c_n > 0$ for every n and $\mathrm{LIM}\, c_n \neq 0$. Prove that $\mathrm{LIM}\, c_n$ is positive.

Solution. (a) For $m, n \geq N$: $|a_m - a_n| \leq \frac{1}{m} + \frac{1}{n} \leq \frac{2}{N} \leq \varepsilon$ once $N \geq 2/\varepsilon$, so (a_n) is Cauchy. Also $|a_n - \frac{1}{2}| = \frac{1}{n} \leq \varepsilon$ for $n \geq 1/\varepsilon$, so (a_n) is equivalent to the constant sequence $\frac{1}{2}$, i.e. $\mathrm{LIM}\, a_n = \frac{1}{2}$.

(b) Put $b_n = \frac{1}{2}$ for $n \leq 3$ and $b_n = a_n$ for $n \geq 4$. For $n \geq 4$, $b_n = a_n \geq \frac{1}{2} - \frac{1}{n} \geq \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$, and $b_n = \frac{1}{2} \geq \frac{1}{4}$ for $n \leq 3$. So $b_n \geq \frac{1}{4} > 0$ for all n, and (b_n) is Cauchy (it is (a_n) with three terms changed). Since $a_n = b_n$ for all $n \geq 4$, we have $|a_n - b_n| = 0 \leq \varepsilon$ for $n \geq 4$ and every ε, so the sequences are equivalent. Hence $\mathrm{LIM}\, a_n = \mathrm{LIM}\, b_n$, which is positive by definition.

(c) Two Cauchy sequences that agree from some finite index onward are equivalent, so they define the same real number. Hence positivity (and the value of LIM) depends only on the tail of the sequence.

★ **KEY METHOD** *(L12)*: **Overwrite finitely many terms.** To obtain a sequence with a nicer property (bounded away from zero, nonnegative, etc.), replace the first $N-1$ terms by a suitable constant. The new sequence is equivalent to the old one, so it has the same LIM. This is how the lecture proves trichotomy of the reals.

(d) Let $x = \mathrm{LIM}\, c_n \neq 0$. Then (c_n) is *not* equivalent to the zero sequence. Negating "for every $\varepsilon > 0$ there is N with $|c_n| \leq \varepsilon$ for all $n \geq N$" gives:

$$\exists\, \varepsilon_0 > 0 \;\; \forall N \;\; \exists\, n \geq N : \; |c_n| > \varepsilon_0, \qquad \text{i.e. } c_n > \varepsilon_0 \text{ (as } c_n > 0).$$

Since (c_n) is Cauchy, there is N_0 with $|c_m - c_n| \leq \frac{\varepsilon_0}{2}$ for all $m, n \geq N_0$. Apply the negation

with $N = N_0$ to get $n_0 \geq N_0$ with $c_{n_0} > \varepsilon_0$. Then for every $m \geq N_0$,

$$c_m \geq c_{n_0} - |c_{n_0} - c_m| > \varepsilon_0 - \tfrac{\varepsilon_0}{2} = \tfrac{\varepsilon_0}{2}.$$

Now overwrite the first $N_0 - 1$ terms by $\frac{\varepsilon_0}{2}$: the new sequence (d_n) is equivalent to (c_n) and $d_n \geq \frac{\varepsilon_0}{2} > 0$ for all n. So $x = \mathrm{LIM}\, d_n$ is positive.

★ **KEY METHOD** *(L12)*: **Negate a $\forall\exists$ statement, then let Cauchy trap the tail.** (i) "Not equivalent to 0" becomes "$\exists\varepsilon_0\ \forall N\ \exists n \geq N$ with $|c_n| > \varepsilon_0$": flip each quantifier and the inequality, keep the order. (ii) One far-out term c_{n_0} together with the Cauchy condition at level $\varepsilon_0/2$ forces *all* later terms into the window $c_{n_0} \pm \frac{\varepsilon_0}{2}$. (iii) Then overwrite finitely many terms. This three-step combination is the proof that a nonzero real is represented by a sequence bounded away from zero, and similar tail arguments recur in exams.

Problem 2. *The inverse of a real number* [L12] ⋆⋆

Let $x = \mathrm{LIM}\, a_n$ where $a_n = \dfrac{n+1}{2n}$.

(a) Show (a_n) is Cauchy and bounded away from zero (find a rational $c > 0$ with $|a_n| \geq c$ for all n).

(b) Show that $c_n = 1/a_n$ is Cauchy and find $\mathrm{LIM}\, c_n$ by proving (c_n) is equivalent to a constant sequence.

(c) Verify directly from the definition of multiplication that $x \cdot \mathrm{LIM}\, c_n = 1$.

Solution. (a) $a_n = \frac{1}{2} + \frac{1}{2n}$, so $|a_m - a_n| = \frac{1}{2}\left|\frac{1}{m} - \frac{1}{n}\right| \leq \frac{1}{2N} \leq \varepsilon$ for $m, n \geq N \geq \frac{1}{2\varepsilon}$. Also $a_n \geq \frac{1}{2}$ for all n, so $c = \frac{1}{2}$ works.

(b) $c_n = \dfrac{2n}{n+1} = 2 - \dfrac{2}{n+1}$. Then $|c_m - c_n| = 2\left|\frac{1}{m+1} - \frac{1}{n+1}\right| \leq \frac{2}{N+1} \leq \varepsilon$ for $m, n \geq N \geq 2/\varepsilon$, so (c_n) is Cauchy. Moreover $|c_n - 2| = \frac{2}{n+1} \leq \varepsilon$ for $n \geq 2/\varepsilon$, so (c_n) is equivalent to the constant sequence 2, and $\mathrm{LIM}\, c_n = 2$.

(c) By definition, $x \cdot \mathrm{LIM}\, c_n = \mathrm{LIM}(a_n c_n) = \mathrm{LIM}(1, 1, 1, \dots) = 1$ since $a_n c_n = 1$ for every n. Hence $x^{-1} = \mathrm{LIM}\, c_n = 2$. (Consistently, $x = \frac{1}{2}$ since $|a_n - \frac{1}{2}| = \frac{1}{2n} \to 0$.)

★ **KEY METHOD** *(L12)*: **Reciprocal of a Cauchy sequence: rescale ε.** For a Cauchy sequence with $|a_n| \geq c > 0$,

$$\left|\frac{1}{a_m} - \frac{1}{a_n}\right| = \frac{|a_n - a_m|}{|a_m||a_n|} \leq \frac{|a_n - a_m|}{c^2}.$$

So apply the Cauchy condition of (a_n) with $\delta = \varepsilon c^2$. Whenever a bound has the shape $C \cdot |a_n - a_m|$, choose the Cauchy tolerance as ε/C. This "rescale ε by the constant" idea is used again in the product law and in the quotient law.

Problem 3. *Comparison, and $\frac{1}{n}$-closeness* [L12] ⋆⋆

(a) Let (a_n) be a Cauchy sequence of rationals with $a_n \geq -\frac{1}{n}$ for all n. Prove $\mathrm{LIM}\, a_n \geq 0$.

(b) Let $(a_n), (b_n)$ be Cauchy sequences of rationals with $|a_n - b_n| \leq \frac{1}{n}$ for all n. Using part (a) only (not the definition of equivalence), prove $\mathrm{LIM}\, a_n = \mathrm{LIM}\, b_n$.

(c) Let $s_n = \sum_{k=1}^{n} \dfrac{1}{k(k+1)}$. Show $s_n = 1 - \frac{1}{n+1}$ and compute $\operatorname{LIM} s_n$.

(d) Let x be a real with $x \geq 0$ and $x \leq \frac{1}{n}$ for every $n \geq 1$. Prove $x = 0$. Deduce: if $x \leq y + \frac{1}{n}$ for every $n \geq 1$ then $x \leq y$.

Solution. (a) $\operatorname{LIM}(-\frac{1}{n}) = 0$ since $(-\frac{1}{n})$ is equivalent to the zero sequence. By the comparison principle (if $a_n \geq b_n$ for all n then $\operatorname{LIM} a_n \geq \operatorname{LIM} b_n$), $\operatorname{LIM} a_n \geq \operatorname{LIM}(-\frac{1}{n}) = 0$.

(b) From $|a_n - b_n| \leq \frac{1}{n}$ we get $a_n \geq b_n - \frac{1}{n}$ for all n, hence $\operatorname{LIM} a_n \geq \operatorname{LIM} b_n - \operatorname{LIM} \frac{1}{n} = \operatorname{LIM} b_n$. Interchanging the roles, $b_n \geq a_n - \frac{1}{n}$ gives $\operatorname{LIM} b_n \geq \operatorname{LIM} a_n$. Together (antisymmetry of $\geq$) the two reals are equal. (Part (a) is the case $b_n = 0$ of the comparison.)

★ **KEY PROOF TECHNIQUE**: **Show $x = y$ by proving $x \leq y$ and $y \leq x$.** Equality of two numbers is almost always proved as two inequalities (antisymmetry). The same idea, applied to *sets*, is proving $A = B$ by showing $A \subseteq B$ and $B \subseteq A$ (see Problems 5 and 7).

★ **KEY METHOD** *(L12)*: **Reduce to a special case via the difference.** The lecture proves "$a_n \geq b_n \Rightarrow \operatorname{LIM} a_n \geq \operatorname{LIM} b_n$" by setting $y_n = a_n - b_n \geq 0$ and applying the special case "nonnegative sequences have nonnegative LIM". In (b) we did the same: the general statement was reduced to (a). Also remember the caution from lecture: $a_n > b_n$ for all n does *not* give $\operatorname{LIM} a_n > \operatorname{LIM} b_n$ (take $a_n = \frac{1}{n}$, $b_n = 0$).

(c) $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$, so the sum telescopes to $1 - \frac{1}{n+1}$. Then $|s_n - 1| = \frac{1}{n+1} \leq \frac{1}{n}$, so by (b) with $b_n = 1$, $\operatorname{LIM} s_n = 1$.

(d) Suppose $x > 0$. By the Archimedean property (with $\varepsilon = x$ and the number 1) there is $n \geq 1$ with $nx > 1$, i.e. $x > \frac{1}{n}$, contradicting $x \leq \frac{1}{n}$. So $x \leq 0$, and with $x \geq 0$ we get $x = 0$. For the deduction: if $x \leq y + \frac{1}{n}$ for all n and $x > y$, then $0 < x - y \leq \frac{1}{n}$ for all n, which contradicts what we just proved. So $x \leq y$.

★ **KEY PROOF TECHNIQUE**: **Prove "$x \leq y$" by contradiction using the Archimedean property.** To show a real $x \leq 0$ (or $x \leq y$), assume $x > 0$, then produce n with $nx > 1$, i.e. $x > \frac{1}{n}$. This converts "smaller than every $\frac{1}{n}$" into "≤ 0" and is used constantly to show two numbers are equal.

Problem 4. *Integer parts* [L13] ⋆⋆

For a real x let $\lfloor x \rfloor$ denote the unique integer m with $m \leq x < m + 1$. Prove:

(a) $\lfloor x + k \rfloor = \lfloor x \rfloor + k$ for every integer k.

(b) $\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1$.

(c) If x is not an integer then $\lfloor -x \rfloor = -\lfloor x \rfloor - 1$.

(d) For every $n \geq 1$, $0 \leq x - \dfrac{\lfloor nx \rfloor}{n} < \dfrac{1}{n}$.

(e) (Existence of the integer part.) Prove by induction on $n \in \{0, 1, 2, \dots\}$ the statement $P(n)$: "for every real x with $0 \leq x < n$ there is an integer m with $m \leq x < m + 1$". Conclude that every real $x \geq 0$ has an integer part.

Solution. We use: an integer m equals $\lfloor z \rfloor$ if and only if $m \leq z < m + 1$.

(a) $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$ gives $\lfloor x \rfloor + k \leq x + k < \lfloor x \rfloor + k + 1$, and $\lfloor x \rfloor + k$ is an integer, so it

equals $\lfloor x+k \rfloor$.

(b) Adding $\lfloor x \rfloor \le x < \lfloor x \rfloor + 1$ and $\lfloor y \rfloor \le y < \lfloor y \rfloor + 1$ gives $\lfloor x \rfloor + \lfloor y \rfloor \le x+y < \lfloor x \rfloor + \lfloor y \rfloor + 2$. The integer $\lfloor x \rfloor + \lfloor y \rfloor$ is $\le x+y$, hence $\le \lfloor x+y \rfloor$ (the floor is the largest integer $\le x+y$, since $\lfloor x+y \rfloor + 1 > x+y$). The integer $\lfloor x+y \rfloor \le x+y < \lfloor x \rfloor + \lfloor y \rfloor + 2$ is a strictly smaller integer than $\lfloor x \rfloor + \lfloor y \rfloor + 2$, hence $\le \lfloor x \rfloor + \lfloor y \rfloor + 1$.

(c) With $m = \lfloor x \rfloor$ we have $m \le x < m+1$ and $x \ne m$ because x is not an integer, so $m < x < m+1$. Negating, $-m-1 < -x < -m$, so $-m-1 \le -x < (-m-1)+1$, i.e. $\lfloor -x \rfloor = -m-1$.

★ **KEY PROOF TECHNIQUE**: **Identify a number by its defining inequalities.** To prove an integer equals $\lfloor z \rfloor$, you do not compute anything: you verify the defining inequalities $m \le z < m+1$ and invoke uniqueness. Use this whenever a "unique number with property P" is defined.

(d) $\lfloor nx \rfloor \le nx < \lfloor nx \rfloor + 1$. Divide by $n > 0$: $\frac{\lfloor nx \rfloor}{n} \le x < \frac{\lfloor nx \rfloor}{n} + \frac{1}{n}$.

★ **KEY METHOD** *(L13)*: **Scale by n, then bracket by integers.** The lecture proves density of $\mathbb{Q}$ and the existence of the least upper bound by multiplying by a large integer N and bracketing Nx between consecutive integers $M-1 \le Nx < M$. Then M/N is a rational within $\frac{1}{N}$ of x. Part (d) is exactly this: $\lfloor nx \rfloor / n$ approximates x to within $\frac{1}{n}$.

(e) *Base case* $n = 0$. $P(0)$ says: if $0 \le x < 0$ then there is an integer m with $m \le x < m+1$. The hypothesis $0 \le x < 0$ is never satisfied, so $P(0)$ is true (vacuously).

Inductive step. Assume $P(n)$ and let $0 \le x < n+1$. If $x < n$, then $P(n)$ gives the required integer. Otherwise $n \le x < n+1$ and $m = n$ works. So $P(n+1)$ holds.

Conclusion. If $x \ge 0$ then by the Archimedean property there is n with $n > x$, and $P(n)$ applies.

★ **KEY METHOD** *(L13)*: **Induction with a vacuous base case.** The lecture proves "every real is bracketed by integers" by induction on a statement of the shape "*if* $z < -M+n$ *then* $\exists m \ldots$". The base case is true *vacuously* because the "if" part is false. Recognise this pattern: when a statement starts with "if $x <$ (small number)", the smallest case costs nothing and the work is in the step.

Problem 5. *A real is the supremum of the rationals below it* [L13] ⋆⋆

Let x be a real number and let $S_x = \{q \in \mathbb{Q} : q < x\}$. You may use that between any two reals there is a rational.

(a) Prove S_x is nonempty and bounded above, and that $\sup S_x = x$.

(b) Prove S_x has no largest element, even when x is rational.

(c) Prove: for reals x, y, $S_x \subseteq S_y \iff x \le y$.

(d) Deduce: $S_x = S_y \iff x = y$.

Solution. (a) By density there is a rational q with $x-1 < q < x$, so $q \in S_x$. By definition x is an upper bound. Suppose $u < x$ were an upper bound of S_x. By density there is a rational q with $u < q < x$; then $q \in S_x$ and $q > u$, contradicting that u is an upper bound. So no upper bound is smaller than x, hence $\sup S_x = x$.

★ **KEY METHOD** *(L13, L14)*: **Show a candidate is the least upper bound: no smaller number is an upper bound.** To prove $s = \sup E$: (1) s is an upper bound; (2) take any $u < s$ and exhibit an element of E exceeding u. Here density of $\mathbb{Q}$ produces that element. The lecture uses precisely this "$x - \varepsilon$ is not an upper bound" step in the $\sqrt{2}$ argument.

(b) If $q \in S_x$, then $q < x$, so by density there is a rational q' with $q < q' < x$. Then $q' \in S_x$ and $q' > q$, so q is not the largest element. This works for rational x as well (so $\sup S_x = x$ need not be attained, even though $x \in \mathbb{Q}$).

(c) ($\Leftarrow$) Let $x \le y$ and $q \in S_x$. Then $q < x \le y$, so $q \in S_y$. Hence $S_x \subseteq S_y$.

($\Rightarrow$) We prove the contrapositive: if $x > y$ then $S_x \not\subseteq S_y$. By density choose a rational q with $y < q < x$. Then $q \in S_x$, but $q < y$ is false, so $q \notin S_y$. Hence $S_x \not\subseteq S_y$.

(d) By mutual containment: $S_x = S_y$ means $S_x \subseteq S_y$ and $S_y \subseteq S_x$. By (c) these say $x \le y$ and $y \le x$, i.e. $x = y$. Conversely, if $x = y$ the sets are literally equal.

★ **KEY PROOF TECHNIQUE**: **Proving $A = B$ for sets: mutual containment.** Write $A \subseteq B$ (take an arbitrary $a \in A$ and show $a \in B$), then $B \subseteq A$ (take an arbitrary $b \in B$ and show $b \in A$). Never try to "see" equality in one step. Here the two inclusions turned into the two inequalities $x \le y$ and $y \le x$.

★ **KEY PROOF TECHNIQUE**: **Proving an "if and only if": do both directions and say which is which.** Label the directions ($\Rightarrow$) and ($\Leftarrow$), state what you assume and what you must show in each, and do not assume the conclusion. Often one direction is easy and the other is best done by contrapositive, as in (c).

★ **KEY PROOF TECHNIQUE**: **Contrapositive.** To prove "$P \Rightarrow Q$" it is equivalent to prove "not $Q \Rightarrow$ not P". Use it when "not Q" hands you something concrete to work with (here: $y < x$ gives a rational between them).

Problem 6. *A set of rationals with no least upper bound in $\mathbb{Q}$* [L13] ⋆⋆⋆

Let $E = \{q \in \mathbb{Q} : q^2 < 5\}$.

(a) Prove that no rational number q satisfies $q^2 = 5$.

(b) Show that E is nonempty and that every positive rational r with $r^2 > 5$ is an upper bound of E. Show also that every upper bound of E in $\mathbb{Q}$ is ≥ 2.

(c) For a positive rational r put $s = \dfrac{3r+5}{r+3}$. Show that s is rational and positive, $s^2 - 5 = \dfrac{4(r^2-5)}{(r+3)^2}$, and $s - r = \dfrac{5-r^2}{r+3}$.

(d) Let r be any rational upper bound of E. Show $r^2 > 5$, and construct a smaller rational upper bound. Conclude that E has no least upper bound in $\mathbb{Q}$.

Solution. (a) Suppose $q = a/b$ with integers $a, b \ge 1$ in lowest terms and $a^2 = 5b^2$. Then $5 \mid a^2$, so $5 \mid a$ (as 5 is prime), say $a = 5k$. Then $25k^2 = 5b^2$, i.e. $b^2 = 5k^2$, so $5 \mid b$ as well. This contradicts lowest terms.

(b) $2 \in E$ since $4 < 5$. Let $r > 0$ be rational with $r^2 > 5$, and let $q \in E$. If $q > r$, then $q > r > 0$ and $q^2 > r^2 > 5$, contradicting $q^2 < 5$. So $q \le r$ for every $q \in E$. Finally, an upper

bound u of E must satisfy $u \geq 2$ because $2 \in E$.

(c) Clearly $s > 0$ is rational. Compute

$$s^2 - 5 = \frac{(3r+5)^2 - 5(r+3)^2}{(r+3)^2} = \frac{9r^2 + 30r + 25 - 5r^2 - 30r - 45}{(r+3)^2} = \frac{4(r^2-5)}{(r+3)^2},$$
$$s - r = \frac{3r + 5 - r^2 - 3r}{r+3} = \frac{5 - r^2}{r+3}.$$

(d) Let r be a rational upper bound, so $r \geq 2 > 0$. By (a), $r^2 \neq 5$. Suppose $r^2 < 5$. By (c), $s > r$ and $s^2 < 5$, so $s \in E$ and $s > r$, contradicting that r is an upper bound. Hence $r^2 > 5$. Now by (c), $s < r$ and $s^2 > 5$, so by (b) s is a rational upper bound smaller than r. Thus no rational upper bound is least, and E has no least upper bound in $\mathbb{Q}$.

★ **KEY METHOD** *(L13)*: **No least upper bound: from any upper bound, build a strictly smaller one.** The lecture shows $\{q \in \mathbb{Q} : q^2 < 2\}$ has no least upper bound by taking an arbitrary upper bound r and producing the smaller upper bound $r' = \frac{1}{2}\left(r + \frac{2}{r}\right)$ (note $r'^2 - 2 = \frac{1}{4}\left(r - \frac{2}{r}\right)^2 > 0$ and $r' < r$). The pattern is: (1) every candidate r is an upper bound of the right shape, (2) a formula gives r' with $r' < r$, (3) r' is again an upper bound, (4) so there is no least one. Also remember the lecture's two small facts: the empty set has no least upper bound (every number M is beaten by $M - 1$), and least upper bounds are unique (if M_1, M_2 are both least then $M_1 \geq M_2$ and $M_2 \geq M_1$).

Problem 7. *Sets of upper bounds, and set equalities* [L13, L14] ⋆⋆⋆

For a nonempty set $A \subseteq \mathbb{R}$ that is bounded above, let $U(A) = \{u \in \mathbb{R} : a \leq u \text{ for all } a \in A\}$ be the set of its upper bounds.

(a) Prove $U(A) = [\sup A, \infty)$.

(b) Prove: $s = \sup A$ if and only if (i) $s \in U(A)$ and (ii) for every $\varepsilon > 0$ there is $a \in A$ with $a > s - \varepsilon$.

(c) For nonempty bounded-above sets A, B prove $U(A \cup B) = U(A) \cap U(B)$, and deduce $\sup(A \cup B) = \max\{\sup A, \sup B\}$.

(d) Prove $\{x \in \mathbb{R} : x \leq \frac{1}{n} \text{ for every } n \geq 1\} = (-\infty, 0]$.

Solution. (a) We show both inclusions.

$U(A) \subseteq [\sup A, \infty)$: let $u \in U(A)$. Then u is an upper bound of A, and $\sup A$ is the *least* upper bound, so $\sup A \leq u$, i.e. $u \in [\sup A, \infty)$.

$[\sup A, \infty) \subseteq U(A)$: let $u \geq \sup A$ and $a \in A$. Then $a \leq \sup A \leq u$, so $u \in U(A)$.

(b) ($\Rightarrow$) Assume $s = \sup A$. Then (i) holds by definition. For (ii), suppose it failed for some $\varepsilon > 0$; then every $a \in A$ satisfies $a \leq s - \varepsilon$, so $s - \varepsilon \in U(A)$, and by (a) $s - \varepsilon \geq \sup A = s$, which is absurd.

($\Leftarrow$) Assume (i) and (ii). By (i), $s \in U(A)$, so $s \geq \sup A$ by (a). If $s > \sup A$, put $\varepsilon = s - \sup A > 0$; by (ii) some $a \in A$ has $a > s - \varepsilon = \sup A$, contradicting that $\sup A$ is an upper bound. So $s = \sup A$.

(c) *Set equality.* $U(A \cup B) \subseteq U(A) \cap U(B)$: an upper bound of $A \cup B$ bounds every element of A and every element of B. $U(A) \cap U(B) \subseteq U(A \cup B)$: if u bounds A and bounds B, then any $x \in A \cup B$ lies in A or in B, so $x \leq u$.

Deduction. $A \cup B$ is nonempty and bounded above, so by (a) applied three times,

$$\begin{aligned}[\sup(A\cup B),\infty) &= U(A\cup B) = U(A)\cap U(B)\\ &= [\sup A,\infty)\cap[\sup B,\infty) = [\max\{\sup A,\sup B\},\infty).\end{aligned}$$

Two sets of the form $[t,\infty)$ are equal only if t (their least element) agrees, so $\sup(A\cup B) = \max\{\sup A, \sup B\}$.

(d) Let T denote the left-hand set.

$T \subseteq (-\infty, 0]$: let $x \in T$ and suppose $x > 0$. By the Archimedean property there is n with $nx > 1$, i.e. $x > \frac{1}{n}$, contradicting $x \in T$. So $x \le 0$.

$(-\infty, 0] \subseteq T$: if $x \le 0$ then $x \le 0 < \frac{1}{n}$ for every $n \ge 1$, so $x \in T$.

★ **KEY PROOF TECHNIQUE**: **Set equality by two inclusions: start each with "let $x \in \ldots$".** In (a), (c) and (d) the proofs are exactly: choose an arbitrary element of the left set and show it lies in the right set; then do the reverse. Always say which inclusion you are proving. Pay attention to the quantifiers hidden in the definition of the set: $U(A)$ has a "for all $a \in A$", which is why a single bad a kills membership.

★ **KEY PROOF TECHNIQUE**: **Iff proofs: prove each direction separately.** In (b) the direction ($\Rightarrow$) uses "if (ii) failed, we would get a smaller upper bound" while ($\Leftarrow$) uses "if $s > \sup A$, (ii) with $\varepsilon = s - \sup A$ gives a contradiction". Both directions are proved by contradiction here, but they are logically different statements, and an exam answer must contain both.

★ **KEY METHOD** *(L13, L14, L16)*: **ε-characterisation of the supremum.** This is the workhorse behind almost every supremum and limit proof in lectures: "$s - \varepsilon$ is not an upper bound", so *some element* (or term) exceeds $s - \varepsilon$. It is used for monotone sequences, for lim sup, and in the proof that $\sqrt{2}$ exists.

Problem 8. *Suprema of reciprocals* [L14] ⋆⋆

Let A be a nonempty set of positive reals and let $A^{-1} = \{1/a : a \in A\}$.

(a) If $m = \inf A > 0$, prove A^{-1} is bounded above and $\sup A^{-1} = 1/m$.

(b) If $\inf A = 0$, prove A^{-1} is not bounded above, i.e. $\sup A^{-1} = +\infty$ in $\mathbb{R}^*$.

(c) Use (a) and (b) to find $\sup\left\{\dfrac{n}{n+1} : n \ge 1\right\}$ and $\sup\left\{n + \dfrac{1}{n} : n \ge 1\right\}$.

Solution. (a) For $a \in A$ we have $a \ge m > 0$, so $\frac{1}{a} \le \frac{1}{m}$; thus $\frac{1}{m}$ is an upper bound of A^{-1}. Let $u < \frac{1}{m}$. If $u \le 0$ then any element of A^{-1} (which is positive) exceeds u. If $u > 0$ then $\frac{1}{u} > m = \inf A$, so $\frac{1}{u}$ is not a lower bound of A: there is $a \in A$ with $a < \frac{1}{u}$, i.e. $\frac{1}{a} > u$. So no $u < \frac{1}{m}$ is an upper bound, and $\sup A^{-1} = \frac{1}{m}$.

(b) Let $M > 0$. Since $\inf A = 0 < \frac{1}{M}$, there is $a \in A$ with $a < \frac{1}{M}$, so $\frac{1}{a} > M$. Hence A^{-1} has no real upper bound, and its supremum in $\mathbb{R}^*$ is $+\infty$.

(c) $\frac{n}{n+1}$ is the reciprocal of $\frac{n+1}{n} = 1 + \frac{1}{n}$. The set $A = \{1 + \frac{1}{n}\}$ has $\inf A = 1$: all elements exceed 1, and given $\varepsilon > 0$ some $1 + \frac{1}{n} < 1 + \varepsilon$ (take $n > 1/\varepsilon$). By (a), $\sup\{\frac{n}{n+1}\} = 1$. Next, $n + \frac{1}{n}$ is the reciprocal of $\frac{n}{n^2+1}$, and $0 < \frac{n}{n^2+1} \le \frac{1}{n}$ shows $\inf\{\frac{n}{n^2+1}\} = 0$. By (b), $\sup\{n + \frac{1}{n}\} = +\infty$.

★ **KEY METHOD** *(L14, L17)*: **Use the defining property of** inf**: a number above** inf A **is not a lower bound.** If $t > \inf A$ then t is not a lower bound of A, so *some element of A is below t.* This one-line fact (and its mirror for sup) is used in (a), (b) and again in the lim sup lemmas of Lecture 17.

Problem 9. *Existence of $\sqrt[3]{2}$ via the least upper bound property* [L14] ⋆⋆⋆

Let $S = \{x \in \mathbb{R} : x \geq 0,\ x^3 < 2\}$.

(a) Show S is nonempty and bounded above by 2, so $s = \sup S$ exists with $1 \leq s \leq 2$.

(b) Suppose $s^3 < 2$. Let $\delta = \min\{1, \frac{2-s^3}{20}\}$ and show $(s+\delta)^3 < 2$. Derive a contradiction.

(c) Suppose $s^3 > 2$. Let $\delta = \dfrac{s^3-2}{6s^2}$. Show $0 < \delta < s$ and $(s-\delta)^3 > 2$, and deduce that $s - \delta$ is an upper bound of S. Derive a contradiction.

(d) Conclude $s^3 = 2$, and show that s is the only positive real with cube equal to 2.

Hint for (c): for $0 \leq \delta \leq s$, $(s-\delta)^3 = s^3 - 3s^2\delta + \delta^2(3s-\delta) \geq s^3 - 3s^2\delta$.

Solution. (a) $1 \in S$. If $x \geq 2$ then $x^3 \geq 8 > 2$, so every element of S is < 2. The least upper bound property gives $s = \sup S$, and $1 \leq s \leq 2$.

(b) With $s \leq 2$ and $0 < \delta \leq 1$ (so $\delta^2 \leq \delta$, $\delta^3 \leq \delta$):

$$(s+\delta)^3 = s^3 + 3s^2\delta + 3s\delta^2 + \delta^3 \leq s^3 + \delta(3s^2 + 3s + 1) \leq s^3 + 19\delta.$$

Since $19\delta \leq \frac{19}{20}(2 - s^3) < 2 - s^3$, we get $(s+\delta)^3 < 2$. Then $s + \delta \geq 0$ lies in S but exceeds s, contradicting that s is an upper bound.

(c) $\delta = \frac{s^3-2}{6s^2} > 0$, and $\delta < \frac{s^3}{6s^2} = \frac{s}{6} < s$. By the hint, $(s-\delta)^3 \geq s^3 - 3s^2\delta = s^3 - \frac{s^3-2}{2} = \frac{s^3+2}{2} > 2$. Now let $x \in S$. If $x \geq s - \delta > 0$ then $x^3 \geq (s-\delta)^3 > 2$, contradicting $x \in S$. So every $x \in S$ satisfies $x < s - \delta$, so $s - \delta$ is an upper bound smaller than s, a contradiction.

(d) By trichotomy only $s^3 = 2$ remains. For uniqueness, if $t > 0$ and $t^3 = 2$ with $t \neq s$, say $t < s$ (the other case is symmetric), then $0 = s^3 - t^3 = (s-t)(s^2 + st + t^2) > 0$, a contradiction.

★ **KEY METHOD** *(L14)*: **Existence by the sup of a test set: rule out $<$ and $>$ with an explicit ε.** Define $E = \{y : y \geq 0,\ y^n \leq \text{target}\}$, find an upper bound (the lecture splits $x \leq 1$ and $x > 1$), take $x = \sup E$, then exclude $x^2 < 2$ and $x^2 > 2$. *Case $x^2 < 2$:* find ε (written in terms of the gap $2 - x^2$, with $0 < \varepsilon < 1$ so that $\varepsilon^2 \leq \varepsilon$) such that $(x+\varepsilon)^2 < 2$; then $x + \varepsilon \in E$ beats the supremum. *Case $x^2 > 2$:* find ε in terms of the gap $x^2 - 2$ such that $(x-\varepsilon)^2 > 2$; then show *by contradiction* that $x - \varepsilon$ is an upper bound (if some $y \in E$ had $y > x - \varepsilon$ then $y^2 > (x-\varepsilon)^2 > 2$). The only creative step is the choice of ε as a fraction of the gap; first do rough bounds like $(x+\varepsilon)^2 \leq x^2 + 5\varepsilon$, then choose ε to make the right side less than 2.

Part B: Sequences, limits and monotone sequences (Lectures 15 and 16)

Problem 10. *Limits straight from the definition* [L15] ⋆

(a) Prove from the definition (find N in terms of ε) that $\dfrac{5n-3}{2n+7} \to \dfrac{5}{2}$.

(b) Prove from the definition that $\sqrt{n^2+1} - n \to 0$.

(c) Let $a_n \geq 0$ for all n and $a_n \to L$. Prove $L \geq 0$.

(d) Let $a_n \to L$ and $b_n \to L'$ with $a_n \leq b_n$ for all n. Prove $L \leq L'$. (*Hint:* suppose $L > L'$ and take $\varepsilon = \frac{L-L'}{3}$.)

(e) True or false: if $x_n < y_n$ for all n and both sequences converge, then $\lim x_n < \lim y_n$. Justify.

Solution. (a) $\left|\dfrac{5n-3}{2n+7} - \dfrac{5}{2}\right| = \dfrac{|10n-6-10n-35|}{2(2n+7)} = \dfrac{41}{2(2n+7)} < \dfrac{41}{4n} \leq \varepsilon$ whenever $n \geq \frac{41}{4\varepsilon}$.

(b) $\sqrt{n^2+1} - n = \dfrac{1}{\sqrt{n^2+1}+n} \leq \dfrac{1}{2n} \leq \varepsilon$ whenever $n \geq \frac{1}{2\varepsilon}$, since $\sqrt{n^2+1} \geq n$.

> ★ **KEY METHOD** *(L15)*: **Finding N: simplify $|a_n - L|$ to something like C/n, then solve $C/n \leq \varepsilon$.** The lecture's example $\frac{1}{n} \to 0$ uses exactly this: bound $|a_n - L|$ by a simple decreasing expression (drop positive terms from denominators, rationalise differences of square roots) and then pick $N \geq C/\varepsilon$, which exists by the Archimedean property.

(c) Suppose $L < 0$ and take $\varepsilon = -L/2 > 0$. There is N with $|a_n - L| \leq -\frac{L}{2}$ for $n \geq N$, so $a_N \leq L - \frac{L}{2} = \frac{L}{2} < 0$, contradicting $a_N \geq 0$. Hence $L \geq 0$.

(d) Suppose $L > L'$ and put $\varepsilon = \frac{L-L'}{3} > 0$. Choose N so that for $n \geq N$ both $|a_n - L| \leq \varepsilon$ and $|b_n - L'| \leq \varepsilon$. Then $a_n \geq L - \varepsilon$ and $b_n \leq L' + \varepsilon$, so

$$b_n - a_n \leq L' - L + 2\varepsilon = -\tfrac{L-L'}{3} < 0,$$

contradicting $a_n \leq b_n$.

> ★ **KEY METHOD** *(L15, L16)*: **Choose ε as a fraction of the gap (and take $N = \max\{N_1, N_2\}$).** When two things must be separated (two supposed limits $L \neq L'$, two supposed limit points, $L > L'$ in (d)), set $\varepsilon = \frac{|L-L'|}{3}$. Then the ε-neighbourhoods are disjoint, and the triangle inequality gives the contradiction $|L - L'| \leq 2\varepsilon < |L - L'|$. The index $N = \max\{N_1, N_2\}$ makes both conditions hold simultaneously. This is the proof that limits are unique, and it reappears in many exam problems.

(e) **False.** $x_n = 0 < \frac{1}{n} = y_n$ for all n, yet both limits equal 0. Strict inequalities only survive as $\lim x_n \leq \lim y_n$ (compare part (d)).

Problem 11. *Averaging consecutive terms* [L15, L16] ⋆⋆

Let $b_n = \dfrac{a_n + a_{n+1}}{2}$.

(a) If $a_n \to L$, prove $b_n \to L$.

(b) Show the converse is false by giving a divergent (a_n) for which (b_n) converges.

(c) If (a_n) is *increasing* and (b_n) converges to L, prove (a_n) converges to L.

Solution. (a) Given $\varepsilon > 0$ choose N with $|a_n - L| \leq \varepsilon$ for all $n \geq N$. For $n \geq N$ both a_n and a_{n+1} are within ε of L, so $|b_n - L| \leq \frac{|a_n - L| + |a_{n+1} - L|}{2} \leq \varepsilon$.

★ **KEY METHOD** *(L15)*: **Rewrite the difference so the triangle inequality applies.** Write $b_n - L = \frac{1}{2}((a_n - L) + (a_{n+1} - L))$, i.e. express the quantity you want to control as a combination of quantities you already control. This is the same "add and subtract" trick used in the lecture proofs of the sum, product and "convergent ⇒ Cauchy" results.

(b) $a_n = (-1)^n$ diverges, but $b_n = 0$ for all n.

(c) Since (a_n) is increasing, $a_n \le b_n \le a_{n+1}$. A convergent sequence is bounded, so $b_n \le M$ for some M, hence $a_n \le M$ for all n. Thus (a_n) is increasing and bounded above, so it converges, say to A (bounded monotone sequences converge). Then $a_{n+1} \to A$ as well, and the limit laws give $b_n \to \frac{A+A}{2} = A$. Limits are unique, so $A = L$.

★ **KEY METHOD** *(L15, L16)*: **Prove convergence first, then identify the limit using uniqueness.** When you do not know a sequence converges you may not take limits of an equation. First prove convergence by a theorem (here: bounded and monotone), and only then pass to the limit; shifted sequences a_{n+1} have the same limit as a_n.

Problem 12. *Two series-like sequences: one Cauchy, one not* [L15, L18] ⋆⋆

(a) Let $a_n = \sum_{k=1}^{n} \dfrac{1}{\sqrt{k}}$. Prove $|a_{2n} - a_n| \ge \frac{1}{2}$ for every n, and conclude that (a_n) is not Cauchy and hence does not converge.

(b) Let $b_n = \sum_{k=1}^{n} \dfrac{1}{k^2}$. Prove that for $m > n \ge 1$, $|b_m - b_n| < \frac{1}{n}$, and conclude that (b_n) converges.

(c) Show that $1 \le \lim b_n \le 2$.

(d) True or false: if (a_n) is a Cauchy sequence of rationals and $x = \mathrm{LIM}\, a_n$, then (a_n) converges to x as a sequence of reals.

Hint for (b): for $k \ge 2$, $\frac{1}{k^2} \le \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$.

Solution. (a) $a_{2n} - a_n = \sum_{k=n+1}^{2n} \frac{1}{\sqrt{k}} \ge n \cdot \frac{1}{\sqrt{2n}} = \sqrt{n/2} \ge \frac{1}{\sqrt{2}} > \frac{1}{2}$. If (a_n) were Cauchy, then with $\varepsilon = \frac{1}{2}$ there would be N such that $|a_m - a_n| \le \frac{1}{2}$ for all $m, n \ge N$; taking $n = N$, $m = 2N$ gives a contradiction. Since convergent sequences are Cauchy, (a_n) does not converge.

★ **KEY METHOD** *(L16)*: **Disprove convergence: show not Cauchy, using a fixed ε and two indices that depend on N.** "Convergent ⇒ Cauchy" has the contrapositive "not Cauchy ⇒ not convergent". To show *not Cauchy*, negate the definition: $\exists \varepsilon\ \forall N\ \exists m, n \ge N$ with $|a_m - a_n| > \varepsilon$. Pick one explicit ε (the lecture takes $\varepsilon = 1$ for $(-1)^n(1 + 10^{-n})$ and shows $|a_n - a_{n+1}| > 2$) and give m, n in terms of N (here $n = N$, $m = 2N$).

(b) For $m > n \ge 1$ (so every k in the sum is at least 2):

$$0 < b_m - b_n = \sum_{k=n+1}^{m} \frac{1}{k^2} \le \sum_{k=n+1}^{m} \left(\frac{1}{k-1} - \frac{1}{k} \right) = \frac{1}{n} - \frac{1}{m} < \frac{1}{n}.$$

Given $\varepsilon > 0$ take $N \ge 1/\varepsilon$; then $|b_m - b_n| < \frac{1}{N} \le \varepsilon$ for $m, n \ge N$. So (b_n) is Cauchy, hence convergent (completeness of $\mathbb{R}$).

(c) b_n is increasing with $b_1 = 1$, so $\lim b_n \ge 1$. Also $b_n \le 1 + \sum_{k=2}^{n} \left(\frac{1}{k-1} - \frac{1}{k} \right) = 2 - \frac{1}{n} < 2$, so $\lim b_n \le 2$.

(d) **True.** This is the lecture result "formal limits are actual limits": $\mathrm{LIM}_{n\to\infty} a_n = \lim_{n\to\infty} a_n$.

Problem 13. *Limit laws: a trap, computations, and a proof from scratch* [L15] ⋆⋆

(a) A student argues: "$n\cdot\frac{1}{n}$: $\lim n$ does not exist, so $\lim n\cdot\frac{1}{n}$ does not exist." Explain precisely what is wrong, and give the correct limit.

(b) Compute $\lim_{n\to\infty}\left(\sqrt{4n^2+n}-2n\right)$.

(c) Compute $\lim_{n\to\infty}\dfrac{1+2+\cdots+n}{n^2}$.

(d) Compute $\lim_{n\to\infty}\dfrac{(n+1)(2n+3)}{n^2+5}$.

(e) Using only the ε-N definition, prove that if $a_n\to a$ then $a_n^3\to a^3$. (You will need that (a_n) is bounded: prove that as well.)

(f) True or false: if $a_n^2\to 4$, then $a_n\to 2$ or $a_n\to -2$.

Solution. (a) The product law says: *if* $a_n\to A$ and $b_n\to B$ then $a_nb_n\to AB$. When one of the limits fails to exist, the law says nothing: it does not claim the product diverges. Here $n\cdot\frac{1}{n}=1$ for every n, so the limit is 1.

(b) $\sqrt{4n^2+n}-2n = \dfrac{n}{\sqrt{4n^2+n}+2n} = \dfrac{1}{\sqrt{4+\frac{1}{n}}+2}$. Since $0\le \sqrt{4+\frac{1}{n}}-2 = \dfrac{1/n}{\sqrt{4+1/n}+2}\le\frac{1}{4n}$, we have $\sqrt{4+\frac{1}{n}}\to 2$ by the squeeze test, so by the quotient law the limit is $\frac{1}{2+2}=\frac{1}{4}$.

(c) $1+2+\cdots+n=\frac{n(n+1)}{2}$, so the quotient is $\frac{n+1}{2n}=\frac{1}{2}+\frac{1}{2n}\to\frac{1}{2}$.

(d) Divide by n^2: $\dfrac{(1+\frac{1}{n})(2+\frac{3}{n})}{1+\frac{5}{n^2}}\to\dfrac{(1+0)(2+0)}{1+0}=2$, using the sum, product and quotient laws (the denominator tends to $1\neq 0$).

(e) *Boundedness.* Take $\varepsilon=1$: there is N with $|a_n-a|\le 1$, hence $|a_n|\le|a|+1$, for $n\ge N$. The finitely many terms $a_1,\dots,a_{N-1}$ satisfy $|a_i|\le M_0$ for some M_0. Put $M=\max\{M_0,|a|+1\}$; then $|a_n|\le M$ for all n and $|a|\le M$.

The estimate. $|a_n^3-a^3|=|a_n-a|\cdot|a_n^2+a_na+a^2|\le 3M^2|a_n-a|$. Given $\varepsilon>0$, choose N' with $|a_n-a|\le\frac{\varepsilon}{3M^2}$ for $n\ge N'$. Then $|a_n^3-a^3|\le\varepsilon$ for $n\ge N'$.

★ **KEY METHOD** *(L15)*: **Convergent ⇒ bounded (use $\varepsilon=1$, then take a max over the finitely many early terms).** After N all terms are within 1 of L, so $|a_n|\le|L|+1$; the first $N-1$ terms form a finite set, hence bounded; take $M=\max\{M_0,|L|+1\}$. You need this bound inside the proof of the product law (it controls the factor that is *not* small).

★ **KEY METHOD** *(L15)*: **Add and subtract / factor, bound the unwanted factor, then split ε.** The product law is proved by $|a_nb_n-xy|\le|a_n-x||b_n|+|b_n-y||x|$, then $\le M_1\varepsilon_1+|x|\varepsilon_2$, then choosing $\varepsilon_1=\frac{\varepsilon}{2M_1}$ and $\varepsilon_2=\frac{\varepsilon}{2(|x|+1)}$ (the "+1" avoids dividing by 0 when $x=0$). In (e) the factorisation does the same job. The recipe: factor out $|a_n-a|$, bound everything else by a constant, choose the tolerance so the total is $\le\varepsilon$.

(f) **False.** $a_n = 2(-1)^n$ has $a_n^2 = 4$ for all n, but (a_n) diverges. (What is true is $|a_n| \to 2$.)

Problem 14. *Supremum and infimum in the extended reals* [L16] ⋆⋆

In $\mathbb{R}^* = \mathbb{R} \cup \{-\infty, +\infty\}$ (ordered by $-\infty < x < +\infty$ for real x) find sup and inf of each set, with brief justification.

(i) $E_1 = \{(-1)^n n : n \geq 1\}$.

(ii) $E_2 = \{\frac{1}{n} : n \geq 1\} \cup \{-\infty\}$.

(iii) $E_3 = \{x \in \mathbb{R} : x < 3\} \cup \{+\infty\}$.

(iv) $E_4 = \varnothing$.

(v) $E_5 = \{x \in \mathbb{R}^* : x \geq 5\}$.

Then prove: *every* subset of $\mathbb{R}^*$ has a supremum in $\mathbb{R}^*$.

Solution. (i) The even terms $2, 4, 6, \ldots$ are unbounded above and the odd terms $-1, -3, \ldots$ are unbounded below, so $\sup E_1 = +\infty$, $\inf E_1 = -\infty$.

(ii) $\max E_2 = 1$ (the largest of $\frac{1}{n}$), and $-\infty \in E_2$ is the smallest element, so $\sup E_2 = 1$, $\inf E_2 = -\infty$.

(iii) $+\infty \in E_3$ is the largest element and E_3 contains arbitrarily negative reals: $\sup E_3 = +\infty$, $\inf E_3 = -\infty$.

(iv) Every element of $\mathbb{R}^*$ is vacuously an upper bound of $\varnothing$, and the least of them is $-\infty$: $\sup \varnothing = -\infty$; similarly $\inf \varnothing = +\infty$.

(v) $E_5 = [5, +\infty]$ contains $+\infty$ and has least element 5: $\sup E_5 = +\infty$, $\inf E_5 = 5$.

Every $E \subseteq \mathbb{R}^*$ has a supremum. Case 1: $+\infty \in E$. Then $+\infty$ is an upper bound and is the least one. Case 2: $E \subseteq \{-\infty\}$ (including $E = \varnothing$). Then $-\infty$ is an upper bound and is the least element of $\mathbb{R}^*$, so $\sup E = -\infty$. Case 3: otherwise $F = E \cap \mathbb{R}$ is nonempty and $+\infty \notin E$. If F is bounded above in $\mathbb{R}$, then $\sup F \in \mathbb{R}$ (least upper bound property) is an upper bound of E (as $-\infty \leq$ everything), and any upper bound u of E in $\mathbb{R}^*$ is $\geq$ every element of F, hence $u \geq \sup F$. So $\sup E = \sup F$. If F is not bounded above in $\mathbb{R}$, then the only upper bound of E in $\mathbb{R}^*$ is $+\infty$, so $\sup E = +\infty$.

★ **KEY PROOF TECHNIQUE**: **Proof by exhaustive cases.** Split into cases that cover *all* possibilities (here: $+\infty \in E$; $E \subseteq \{-\infty\}$; the rest), prove the claim in each case, and check no case is missing. The lecture's definition of sup on $\mathbb{R}^*$ is exactly this case split.

Problem 15. *Monotone convergence for a recursion* [L16] ⋆⋆

Consider $a_{n+1} = \dfrac{1 + a_n^2}{2}$.

(a) Take $a_1 = 0$. Show $0 \leq a_n < 1$ for all n and that (a_n) is increasing. Conclude it converges and find the limit.

(b) Take $a_1 = 2$. Show (a_n) is increasing, and prove that it diverges (so $\sup a_n = +\infty$).

(c) Let (a_n) be *any* increasing sequence and suppose the subsequence (a_{2n}) converges to L. Prove that the whole sequence (a_n) converges to L.

Hint: $a_{n+1} - a_n = \frac{1}{2}(a_n - 1)^2$.

Solution. Note $a_{n+1} - a_n = \frac{1+a_n^2}{2} - a_n = \frac{(a_n-1)^2}{2} \geq 0$ always, so the sequence is increasing whatever a_1 is.

(a) $a_1 = 0 \in [0,1)$. If $0 \leq a_n < 1$ then $0 \leq a_n^2 < 1$ so $\frac{1}{2} \leq a_{n+1} < 1$. Thus $0 \leq a_n < 1$ for all n. A bounded increasing sequence converges, say to L. Also $a_{n+1} \to L$, so the limit laws give $L = \frac{1+L^2}{2}$, i.e. $(L-1)^2 = 0$, so $L = 1$.

★ **KEY METHOD** *(L16)*: **Bounded monotone ⇒ convergent; then pass to the limit in the recursion.** Step 1: prove monotone and bounded (usually by induction). Step 2: the limit L exists, and $a_{n+1} \to L$ too, so $L = f(L)$. Step 3: solve $L = f(L)$ and use the known bound ($L \geq a_1$, $L \leq M$) to pick the right root. The lecture does exactly this for x^n with $0 < x < 1$: from $L = xL$ it gets $L(x-1) = 0$, so $L = 0$.

(b) Since $a_1 = 2 > 1$, $a_{n+1} - a_n = \frac{(a_n-1)^2}{2} > 0$ as long as $a_n \neq 1$, and inductively $a_{n+1} > a_n \geq 2$. Suppose (a_n) converged to L. As in (a), $L = 1$. But $a_n \geq 2$ for all n forces $L \geq 2$ (a limit of numbers ≥ 2 is ≥ 2, by Problem 10(c) applied to $a_n - 2$), a contradiction. So (a_n) diverges. Being increasing and not convergent, it is unbounded above (a bounded increasing sequence converges), so $\sup a_n = +\infty$.

★ **KEY PROOF TECHNIQUE**: **Proof by contradiction: assume the limit exists and derive an impossible value.** In (b) the recursion shows that *if* a limit exists it must be 1; the data $a_n \geq 2$ rules that out. So the assumption "convergent" fails. Note the logic: the algebra only tells you what the limit would have to be.

(c) Let $\varepsilon > 0$. Choose K with $|a_{2k} - L| \leq \varepsilon$ for all $k \geq K$. Let $n \geq 2K$ and put $k = \lfloor n/2 \rfloor \geq K$, so $2k \leq n \leq 2k+2$. Since (a_n) is increasing, $a_{2k} \leq a_n \leq a_{2k+2}$. Both a_{2k} and a_{2k+2} lie in $[L-\varepsilon, L+\varepsilon]$ (the latter has even index $2(k+1)$ with $k+1 \geq K$). Hence $L - \varepsilon \leq a_n \leq L + \varepsilon$ for all $n \geq 2K$, i.e. $a_n \to L$.

★ **KEY METHOD** *(L16)*: **Sandwich a monotone sequence between two terms you control.** The lecture proof of "bounded monotone ⇒ convergent" fixes ε, uses that $S - \varepsilon$ is not an upper bound to get one index N with $a_N > S - \varepsilon$, and then monotonicity propagates the inequality to *all* $n \geq N$, giving $S - \varepsilon < a_n \leq S$. Part (c) is the same idea: monotonicity lets two controlled terms trap every term in between.

Problem 16. *Sup of a sequence versus its limit* [L16] ⋆⋆

Let $a_n = \dfrac{n^2}{2^n}$.

(a) Show $\dfrac{a_{n+1}}{a_n} = \dfrac{(n+1)^2}{2n^2}$. Show this ratio exceeds 1 for $n = 1, 2$ and is at most 1 for all $n \geq 3$.

(b) Deduce $\sup_n a_n = \max_n a_n = \frac{9}{8}$.

(c) Prove $a_n \to 0$, and hence $\inf_n a_n = 0$ (not attained).

Solution. (a) $\dfrac{a_{n+1}}{a_n} = \dfrac{(n+1)^2/2^{n+1}}{n^2/2^n} = \dfrac{(n+1)^2}{2n^2}$. This is ≤ 1 iff $(n+1)^2 \leq 2n^2$ iff $n^2 - 2n - 1 \geq 0$ iff $(n-1)^2 \geq 2$. This fails for $n = 1, 2$ (giving ratios 2 and $\frac{9}{8}$, both > 1) and holds for $n \geq 3$ since $(n-1)^2 \geq 4$.

(b) So $a_1 = \frac{1}{2} < a_2 = 1 < a_3 = \frac{9}{8}$ and $a_3 \geq a_4 \geq a_5 \geq \cdots$. Hence the largest term is $a_3 = \frac{9}{8}$:

$\sup = \max = \frac{9}{8}$.

(c) $(a_n)_{n\geq 3}$ is decreasing and bounded below by 0, so it converges to some $L \geq 0$. Now $a_{n+1} = a_n \cdot r_n$ with $r_n = \frac{(1+1/n)^2}{2} \to \frac{1}{2}$ by the limit laws. Taking limits, $L = L \cdot \frac{1}{2}$, so $L = 0$. All terms are positive and decrease to 0, so $\inf a_n = 0$ and it is not attained. (Note the contrast: $\sup_n a_n = \frac{9}{8}$ but $\lim a_n = 0$.)

★ **KEY METHOD** *(L16)*: **Compare consecutive terms to find monotonicity, then use $L = L \cdot \lim$(ratio).** Dividing a_{n+1} by a_n shows where a sequence turns from increasing to decreasing. Once it is monotone and bounded the limit exists, and passing to the limit in $a_{n+1} = a_n r_n$ with $r_n \to r < 1$ gives $L = rL$, so $L = 0$. This is the lecture's argument for $x^n \to 0$ in disguise.

Part C: limsup, liminf, squeeze, subsequences, BW (Lectures 17 and 18)

Problem 17. *Computing* lim sup, lim inf *and limit points* [L17] ⋆⋆

Recall $a_N^+ = \sup_{n\geq N} a_n$, $a_N^- = \inf_{n\geq N} a_n$, $\limsup a_n = \inf_N a_N^+$, $\liminf a_n = \sup_N a_N^-$. For each sequence find sup, inf, lim sup, lim inf (in $\mathbb{R}^*$) and the set of real limit points.

(i) $a_n = r_n + \frac{1}{n}$, where $r_n \in \{0, \frac{1}{3}, \frac{2}{3}\}$ is $\frac{1}{3}(n \bmod 3)$.

(ii) $b_n = (-1)^n \left(1 - \frac{1}{n}\right)$.

(iii) $c_n = n$ if $3 \mid n$, and $c_n = (-1)^n$ otherwise.

(iv) For the sequence (b_n) in (ii), prove *by two inclusions* that the set of limit points is exactly $\{-1, 1\}$.

Solution. (i) The three families are $a_n = \frac{1}{n}$ $(3 \mid n)$, $\frac{1}{3} + \frac{1}{n}$ $(n \equiv 1)$, $\frac{2}{3} + \frac{1}{n}$ $(n \equiv 2)$; they tend to $0, \frac{1}{3}, \frac{2}{3}$, so the limit points are $\{0, \frac{1}{3}, \frac{2}{3}\}$. Sup: $a_1 = \frac{4}{3}$ is the largest term ($a_2 = \frac{7}{6}$, and all later terms are at most $\frac{2}{3} + \frac{1}{3} < \frac{7}{6}$ for $n \geq 3$), so $\sup = \max = \frac{4}{3}$. Inf: all terms are positive and $a_{3k} = \frac{1}{3k} \to 0$, so $\inf = 0$, not attained. For limsup: for every N, $a_N^+ \leq \frac{2}{3} + \frac{1}{N}$ (as $r_n \leq \frac{2}{3}$), and $a_N^+ \geq \frac{2}{3}$ (there are terms $\frac{2}{3} + \frac{1}{n} > \frac{2}{3}$ with $n \geq N$). So $\frac{2}{3} \leq a_N^+ \leq \frac{2}{3} + \frac{1}{N}$, and $\limsup = \inf_N a_N^+ = \frac{2}{3}$. For liminf: $a_N^- \leq a_{3k} = \frac{1}{3k} \leq \frac{1}{N}$ for multiples of three $3k \geq N$, and $a_N^- > 0$, so $0 < a_N^- \leq \frac{1}{N}$ and $\liminf = \sup_N a_N^- = 0$.

(ii) Even n: $1 - \frac{1}{n} \uparrow 1$ (values $\frac{1}{2}, \frac{3}{4}, \dots$). Odd n: $-(1 - \frac{1}{n}) \downarrow -1$ (values $0, -\frac{2}{3}, \dots$). So $\sup = 1$ and $\inf = -1$, both not attained. Every tail contains even terms approaching 1 and no term is ≥ 1, so $b_N^+ = 1$ for all N and $\limsup = 1$; likewise $b_N^- = -1$ and $\liminf = -1$. Limit points: $\{-1, 1\}$ (proved in (iv)).

(iii) The terms with $3 \mid n$ go to $+\infty$ and contribute no real limit point. The other terms are $(-1)^n$: values -1 $(n = 1)$, $+1$ $(n = 2)$, $+1$ $(n = 4)$, -1 $(n = 5)$, and so on, each appearing infinitely often. Limit points: $\{-1, 1\}$. $\sup = +\infty$; $\inf = -1$ (attained at $n = 1$). Every tail is unbounded above, so $c_N^+ = +\infty$ and $\limsup = +\infty$. Every tail contains -1 and all terms are ≥ -1, so $c_N^- = -1$ and $\liminf = -1$. Note that $\limsup = +\infty$ is not a real limit point, even though 1 is.

★ **KEY METHOD** *(L17)*: **Compute a_N^+ separately for each residue/parity of N, then take the infimum.** For $a_n = (-1)^n(1+10^{-n})$ the lecture splits into even and odd terms, finds $a_N^+ = 1+10^{-2k}$ for $N \in \{2k-1, 2k\}$ (a decreasing sequence), and takes $\inf_N a_N^+ = \lim_N a_N^+ = 1$. Or, as in (i), *squeeze* a_N^+ between two simple bounds (here $\frac{2}{3} \le a_N^+ \le \frac{2}{3} + \frac{1}{N}$) and read off the infimum.

(iv) Let L be the set of limit points of (b_n). Recall: v is a limit point iff $\forall \varepsilon > 0\ \forall N\ \exists n \ge N$ with $|b_n - v| \le \varepsilon$.

$\{-1, 1\} \subseteq L$: Fix $\varepsilon > 0$ and N. Choose an even $n \ge N$ with $\frac{1}{n} \le \varepsilon$; then $|b_n - 1| = \frac{1}{n} \le \varepsilon$, so $1 \in L$. Choose an odd $n \ge N$ with $\frac{1}{n} \le \varepsilon$; then $|b_n + 1| = \frac{1}{n} \le \varepsilon$, so $-1 \in L$.

$L \subseteq \{-1, 1\}$: we prove the contrapositive. Let $v \notin \{-1, 1\}$ and $d = \min\{|v-1|, |v+1|\} > 0$. Take $\varepsilon = \frac{d}{3}$ and $N \ge \frac{3}{d}$. For every $n \ge N$, $|b_n - (-1)^n| = \frac{1}{n} \le \frac{d}{3}$, so

$$|b_n - v| \ge |(-1)^n - v| - |b_n - (-1)^n| \ge d - \tfrac{d}{3} = \tfrac{2d}{3} > \tfrac{d}{3} = \varepsilon.$$

So for this ε and this N no $n \ge N$ has $|b_n - v| \le \varepsilon$: $v \notin L$.

★ **KEY METHOD** *(L16)*: **To show v is NOT a limit point, negate the definition and choose ε as a fraction of a distance.** Negation of "$\forall \varepsilon\ \forall N\ \exists n \ge N :\ |a_n - v| \le \varepsilon$" is "$\exists \varepsilon\ \exists N\ \forall n \ge N :\ |a_n - v| > \varepsilon$". So you must *choose* ε and N. The lecture takes $d = \min\{|x-1|, |x+1|\}$, $\varepsilon = \frac{d}{3}$, N with $Nd \ge 3$, and then uses the triangle inequality $d \le |(-1)^n - x| \le |(-1)^n - a_n| + |a_n - x|$ to show every later term stays away from x.

★ **KEY PROOF TECHNIQUE**: **Describing a set exactly: show $\supseteq$ by exhibiting elements, $\subseteq$ by the contrapositive.** Inclusions of the form "every point not in the set is not a limit point" are best proved by contrapositive.

Problem 18. *Negations, limit points and infinite index sets* [L16, L18] ⋆⋆⋆

(a) Write, using quantifiers, the negation of "$a_n \to L$" and the negation of "L is a limit point of (a_n)".

(b) Prove: $a_n \not\to L$ if and only if there are $\varepsilon > 0$ and a subsequence (a_{n_j}) with $|a_{n_j} - L| > \varepsilon$ for all j.

(c) For a real L and $\varepsilon > 0$ let $I_\varepsilon(L) = \{n \ge 1 : |a_n - L| \le \varepsilon\}$. Prove: L is a limit point of (a_n) if and only if $I_\varepsilon(L)$ is infinite for every $\varepsilon > 0$.

(d) Use (c) to prove: if $a_n \to L$ then L is a limit point and it is the *only* limit point of (a_n).

Solution. (a) $a_n \not\to L$: $\exists \varepsilon > 0\ \ \forall N\ \ \exists n \ge N :\ |a_n - L| > \varepsilon$. And "$L$ is not a limit point": $\exists \varepsilon > 0\ \ \exists N\ \ \forall n \ge N :\ |a_n - L| > \varepsilon$.

★ **KEY METHOD** *(L12, L16)*: **Negating quantifiers.** Swap every $\forall$ with $\exists$, keep the order, and negate the final statement ($\le$ becomes $>$). This is how the lecture shows a sequence is not equivalent to zero, not Cauchy, and that a point is not a limit point. Practise it until automatic: negating "converges to L" and negating "limit point" differ only in the position of N.

(b) ($\Rightarrow$) By (a) there is $\varepsilon > 0$ such that for every N some $n \ge N$ has $|a_n - L| > \varepsilon$. Fix this ε. Choose $n_1 \ge 1$ with $|a_{n_1} - L| > \varepsilon$. Having chosen n_j, apply the statement with $N = n_j + 1$ to

get $n_{j+1} \geq n_j + 1 > n_j$ with $|a_{n_{j+1}} - L| > \varepsilon$. This recursion defines the required subsequence.

($\Leftarrow$) Suppose such ε and (n_j) exist, and suppose for contradiction that $a_n \to L$. Then there is N with $|a_n - L| \leq \varepsilon$ for all $n \geq N$. Since $n_j \geq j$ (by induction from $n_1 \geq 1$ and $n_{j+1} > n_j$), $n_N \geq N$ and so $|a_{n_N} - L| \leq \varepsilon$, contradicting $|a_{n_N} - L| > \varepsilon$.

★ **KEY METHOD** *(L18)*: **Build a subsequence by recursion: "choose the next index larger than the previous one".** To extract a subsequence with a property that holds for infinitely many n, choose n_1, and given n_j apply the property with $N = n_j + 1$ to get $n_{j+1} > n_j$. The lecture does exactly this (with $\varepsilon = \frac{1}{j+1}$ and $N = f(j)+1$) to show that a limit point is the limit of a subsequence, and it needs the fact $f(n) \geq n$ for strictly increasing f, proved by induction.

(c) ($\Rightarrow$) Let L be a limit point and fix $\varepsilon > 0$. Suppose $I_\varepsilon(L)$ were finite. Then it is bounded above by some M (take $M = 0$ if it is empty). Applying the definition of limit point with $N = M + 1$ gives $n \geq M + 1$ with $|a_n - L| \leq \varepsilon$, i.e. $n \in I_\varepsilon(L)$ with $n > M$, a contradiction. So $I_\varepsilon(L)$ is infinite.

($\Leftarrow$) Assume $I_\varepsilon(L)$ is infinite for every $\varepsilon > 0$. Fix ε and N. The finite set $\{1, \dots, N-1\}$ cannot contain the infinite set $I_\varepsilon(L)$, so some $n \in I_\varepsilon(L)$ has $n \geq N$, i.e. $|a_n - L| \leq \varepsilon$.

(d) Let $a_n \to L$ and $\varepsilon > 0$. There is N with $|a_n - L| \leq \varepsilon$ for $n \geq N$, so $I_\varepsilon(L) \supseteq \{N, N+1, \dots\}$ is infinite. By (c), L is a limit point. Now let $L' \neq L$ and put $\varepsilon = \frac{|L-L'|}{3}$. We claim $I_\varepsilon(L') \subseteq \mathbb{N} \setminus I_\varepsilon(L)$: if n were in both sets, then $|L - L'| \leq |L - a_n| + |a_n - L'| \leq 2\varepsilon < |L - L'|$, absurd. Since $\mathbb{N} \setminus I_\varepsilon(L) \subseteq \{1, \dots, N-1\}$ is finite, $I_\varepsilon(L')$ is finite, so L' is not a limit point by (c).

★ **KEY PROOF TECHNIQUE**: **Translating a statement about sequences into a statement about sets of indices.** "$\forall N\ \exists n \geq N$" says exactly that the set of good indices is infinite; "$\exists N\ \forall n \geq N$" says that the set of bad indices is finite. Once translated, set arguments (inclusion, disjointness, finite $\subseteq$ finite) replace ε manipulations.

★ **KEY PROOF TECHNIQUE**: **Iff proofs and set containment together.** Parts (b) and (c) are both "if and only if" statements, each proved in two separate directions. In (d) the key step is a set *inclusion* $I_\varepsilon(L') \subseteq \mathbb{N} \setminus I_\varepsilon(L)$ proved by taking an arbitrary n in the left set.

Problem 19. *Algebra of* lim sup *and* lim inf [L17, L18] ⋆⋆⋆

Let $(a_n), (b_n)$ be bounded real sequences.

(a) Prove $\liminf(-a_n) = -\limsup a_n$.

(b) If $a_n \leq b_n$ for all n, prove $\limsup a_n \leq \limsup b_n$ and $\liminf a_n \leq \liminf b_n$.

(c) If $a_n \to L$, prove $\limsup(a_n + b_n) = L + \limsup b_n$.

(d) Give an example where $\limsup(a_n b_n) \neq \limsup a_n \cdot \limsup b_n$.

(e) Let c be a real number. Prove: (i) if $c > \limsup a_n$ then $a_n < c$ for all large n; (ii) if $c < \limsup a_n$ then for every N there is $n \geq N$ with $a_n > c$.

Fact you may use: for a bounded nonempty set A, $\inf(-A) = -\sup A$.

Solution. (a) $(-a)_N^- = \inf_{n \geq N}(-a_n) = -\sup_{n \geq N} a_n = -a_N^+$. Hence $\liminf(-a_n) = \sup_N(-a_N^+) = -\inf_N a_N^+ = -\limsup a_n$ (using the fact again for the set $\{a_N^+\}$).

(b) $a_n \le b_n$ for all n gives $a_N^+ \le b_N^+$ and $a_N^- \le b_N^-$ for each N. Then $\inf_M a_M^+ \le a_N^+ \le b_N^+$ for every N, so $\inf_M a_M^+ \le \inf_N b_N^+$, i.e. $\limsup a \le \limsup b$. Similarly, $a_N^- \le b_N^- \le \sup_M b_M^-$ for each N, so $\sup_N a_N^- \le \sup_M b_M^-$.

★ **KEY METHOD** *(L18)*: **Comparison of** $\sup / \inf$**: a lower (upper) bound for one set is a lower (upper) bound for the other, so it is** $\le$ **the greatest lower (least upper) bound.** The lecture proves $\inf a_n \le \inf b_n$ for $a_n \le b_n$ as follows: $\inf a_n \le a_n \le b_n$ for all n, so $\inf a_n$ is a lower bound of (b_n), and every lower bound is $\le \inf b_n$. For lim sup, compare the tails $a_N^+ \le b_N^+$ first and then take $\inf_N$.

(c) Fix $\varepsilon > 0$ and choose N_0 with $L - \varepsilon \le a_n \le L + \varepsilon$ for $n \ge N_0$. For $N \ge N_0$, $b_n + L - \varepsilon \le a_n + b_n \le b_n + L + \varepsilon$ for $n \ge N$, so taking suprema over $n \ge N$: $b_N^+ + L - \varepsilon \le (a+b)_N^+ \le b_N^+ + L + \varepsilon$. The tail suprema $(a+b)_N^+$ and b_N^+ are decreasing in N, so their infima over $N \ge N_0$ equal their infima over all N. Taking $\inf_N$: $\limsup b + L - \varepsilon \le \limsup(a+b) \le \limsup b + L + \varepsilon$. Since ε is arbitrary, $\limsup(a_n + b_n) = L + \limsup b_n$.

★ **KEY METHOD** *(L18)*: **"$|x - y| \le \varepsilon$ for every $\varepsilon > 0$ implies $x = y$".** This is the lecture's way to conclude equality of two numbers from approximate equality: if $x \ne y$ take $\varepsilon = \frac{|x-y|}{2}$ to get $|x - y| \le \frac{|x-y|}{2}$, absurd. It closes the proofs of (c) above and of "Cauchy $\Rightarrow$ convergent" ($L^+ - L^- \le 2\varepsilon$ for all ε).

(d) Take $a_n = (-1)^n$ and $b_n = -(-1)^n$. Then $\limsup a_n = \limsup b_n = 1$, but $a_n b_n = -1$ for all n, so $\limsup(a_n b_n) = -1 \ne 1$.

(e) (i) Since $c > \limsup a_n = \inf_N a_N^+$, the number c is *not a lower bound* of $\{a_N^+ : N \ge 1\}$ (it exceeds the greatest lower bound). So some N has $a_N^+ < c$, and then $a_n \le a_N^+ < c$ for all $n \ge N$.

(ii) $\limsup a_n = \inf_N a_N^+ \le a_N^+$ for every N, so $c < a_N^+ = \sup_{n \ge N} a_n$. Thus c is not an upper bound of $\{a_n : n \ge N\}$, so some $n \ge N$ has $a_n > c$.

★ **KEY METHOD** *(L17, L18)*: **If** $t > \inf S$ **then some element of** S **is below** t **(and dually for** sup**).** Statements (i) and (ii) are the lemmas the lecture uses to show that the limsup is a limit point, that $\limsup = \liminf = c$ implies convergence, and in the proof of Bolzano-Weierstrass. The argument is always the same one-liner: a number strictly above the greatest lower bound cannot itself be a lower bound.

Problem 20. *Squeeze test in action* [L18] ⋆⋆

(a) Show $\dfrac{n + \cos n}{n + 1} \to 1$.

(b) Let $y_n = \displaystyle\sum_{k=1}^{n} \frac{n}{n^2 + k}$. Show $y_n \to 1$.

(c) Suppose $|a_{n+1}| \le \frac{3}{4}|a_n|$ for all $n \ge 2$. Prove $a_n \to 0$ (you may use $(\frac{3}{4})^n \to 0$). Apply this to $a_n = \dfrac{n}{2^n}$.

(d) A student "proves" the squeeze test: "If $a_n \le b_n \le c_n$ and $\lim a_n = \lim c_n = \ell$, take limits to get $\ell \le \lim b_n \le \ell$." Identify the gap, and repair the proof in two different ways.

Solution. (a) Since $-1 \le \cos n \le 1$: $\dfrac{n-1}{n+1} \le \dfrac{n + \cos n}{n+1} \le \dfrac{n+1}{n+1} = 1$. The lower bound equals $1 - \frac{2}{n+1} \to 1$ and the upper bound is constant 1. By the squeeze test the limit is 1.

(b) For $1 \le k \le n$: $\frac{n}{n^2+n} \le \frac{n}{n^2+k} \le \frac{n}{n^2+1}$. Summing the n terms, $\dfrac{n^2}{n^2+n} \le y_n \le \dfrac{n^2}{n^2+1}$, i.e. $\dfrac{n}{n+1} \le y_n \le 1 - \dfrac{1}{n^2+1}$. Both bounds tend to 1 (limit laws), so $y_n \to 1$.

(c) By induction, $|a_n| \le \left(\frac{3}{4}\right)^{n-2} |a_2|$ for $n \ge 2$. So $-\left(\frac{3}{4}\right)^{n-2} |a_2| \le a_n \le \left(\frac{3}{4}\right)^{n-2} |a_2|$, and both bounds tend to 0 (a constant multiple of $(\frac{3}{4})^n$). By squeeze, $a_n \to 0$. For $a_n = \frac{n}{2^n}$: $\left|\frac{a_{n+1}}{a_n}\right| = \frac{n+1}{2n} \le \frac{3}{4}$ iff $4n+4 \le 6n$ iff $n \ge 2$, so $\frac{n}{2^n} \to 0$.

(d) *The gap.* The inequality $\lim a_n \le \lim b_n \le \lim c_n$ is only available once we know that $\lim b_n$ *exists*. For all we know at that point (b_n) might diverge, and a limit that has not been shown to exist cannot be compared.

Repair 1 (ε-sandwich). Given $\varepsilon > 0$ choose N_1, N_2 with $\ell - \varepsilon \le a_n$ for $n \ge N_1$ and $c_n \le \ell + \varepsilon$ for $n \ge N_2$. For $n \ge \max\{N_1, N_2\}$: $\ell - \varepsilon \le a_n \le b_n \le c_n \le \ell + \varepsilon$, so $|b_n - \ell| \le \varepsilon$.

Repair 2 ($\limsup$ / $\liminf$). These always exist (for bounded sequences). By the comparison principle, $\ell = \lim a_n = \limsup a_n \le \limsup b_n \le \limsup c_n = \ell$, and similarly $\ell \le \liminf b_n \le \ell$. So $\limsup b_n = \liminf b_n = \ell$ and hence $b_n \to \ell$ (convergent iff $\limsup = \liminf$).

★ **KEY METHOD** *(L18)*: **When the limit is not known to exist, do not take limits: use ε-sandwiches or $\limsup$ / $\liminf$.** The lecture gives three attempts at the squeeze test: the two repairs above work; the naive "take limits" attempt fails because it presupposes that $\lim b_n$ exists. Whenever you want to prove a limit exists, you must use a method that does not assume it: ε-N, Cauchy, monotone bounded, or $\limsup = \liminf$.

Problem 21. *An oscillation criterion, and almost-decreasing sequences* [L17, L18] ⋆⋆⋆

Let (a_n) be a bounded real sequence.

(a) Prove: (a_n) converges if and only if for every $\varepsilon > 0$ there is N with $a_N^+ - a_N^- \le \varepsilon$.

(b) Suppose that for every $\varepsilon > 0$ there is N such that $a_n \le a_m + \varepsilon$ whenever $n \ge m \ge N$. Prove $\limsup a_n \le \liminf a_n + \varepsilon$ for every $\varepsilon > 0$, and deduce that (a_n) converges.

(c) Show that every bounded decreasing sequence satisfies the hypothesis of (b), and that $a_n = \frac{(-1)^n}{n}$ satisfies it too.

You may use: $\liminf a_n \le \limsup a_n$ for bounded sequences, and that a bounded sequence converges if $\limsup a_n = \liminf a_n$.

Solution. (a) ($\Rightarrow$) Suppose $a_n \to L$ and let $\varepsilon > 0$. Choose N with $L - \frac{\varepsilon}{2} \le a_n \le L + \frac{\varepsilon}{2}$ for all $n \ge N$. Then $a_N^+ \le L + \frac{\varepsilon}{2}$ and $a_N^- \ge L - \frac{\varepsilon}{2}$, so $a_N^+ - a_N^- \le \varepsilon$.

($\Leftarrow$) Given $\varepsilon > 0$, take N with $a_N^+ - a_N^- \le \varepsilon$. For $m, n \ge N$ we have $a_N^- \le a_m, a_n \le a_N^+$, so $|a_m - a_n| \le a_N^+ - a_N^- \le \varepsilon$. Hence (a_n) is Cauchy, and therefore convergent (completeness of $\mathbb{R}$).

(b) Fix $\varepsilon > 0$ and N as in the hypothesis. For $m \ge N$, every a_n with $n \ge m$ satisfies $a_n \le a_m + \varepsilon$, so $a_m^+ = \sup_{n \ge m} a_n \le a_m + \varepsilon$. Since $\limsup a_n = \inf_M a_M^+ \le a_m^+$, we get $\limsup a_n \le a_m + \varepsilon$, i.e. $a_m \ge \limsup a_n - \varepsilon$ for *all* $m \ge N$. Therefore $a_N^- = \inf_{m \ge N} a_m \ge \limsup a_n - \varepsilon$. Finally $\liminf a_n = \sup_M a_M^- \ge a_N^-$, so

$$\limsup a_n \le a_N^- + \varepsilon \le \liminf a_n + \varepsilon.$$

As $\varepsilon > 0$ was arbitrary, $\limsup a_n \le \liminf a_n$ (the lecture's "$|x-y| \le \varepsilon$ for all ε" argument), and combined with $\liminf \le \limsup$ we get equality. So (a_n) converges.

(c) If (a_n) is decreasing then $a_n \le a_m$ for $n \ge m$, so the hypothesis holds with $N = 1$ for every ε. For $a_n = \frac{(-1)^n}{n}$ and $n \ge m \ge N$: $a_n \le \frac{1}{n} \le \frac{1}{N}$ and $a_m \ge -\frac{1}{m} \ge -\frac{1}{N}$, so $a_n - a_m \le \frac{2}{N} \le \varepsilon$ if $N \ge \frac{2}{\varepsilon}$. This sequence is not monotone, so the criterion in (b) is strictly more general than "bounded and monotone".

★ **KEY PROOF TECHNIQUE**: **Prove convergence via** $\limsup = \liminf$**.** Once you know $\limsup \le \liminf + \varepsilon$ for all ε, you are done because $\liminf \le \limsup$ always. This is often the easiest way to prove convergence of a sequence given only a one-sided or approximate condition.

★ **KEY METHOD** *(L17)*: **Use** $\limsup \le a_m^+$ **and** $\liminf \ge a_N^-$ **to move between tails and the extremal quantities.** The two inequalities $\inf_M a_M^+ \le a_m^+$ and $a_N^- \le \sup_M a_M^-$ (an infimum is below every member; a supremum is above every member) are the only facts needed to pass between the tail quantities $a_N^\pm$ and $\limsup$ / $\liminf$. The lecture's chain $a_N^- \le a_K \le a_M^+$ with $K = \max\{N, M\}$ (used to prove $\liminf \le \limsup$) is the same idea.

Problem 22. *Subsequences and* $\limsup$ [L18] ⋆⋆⋆

(a) If $a_{2n} \to L$ and $a_{2n-1} \to L$, prove $a_n \to L$.

(b) Give a bounded sequence with $a_{2n} \to L$ for some L such that (a_n) diverges.

(c) Let (a_n) be bounded with $\limsup a_n = 3$ and $\liminf a_n = 1$. Prove: (i) there is a subsequence converging to 3, and one converging to 1; (ii) every convergent subsequence has its limit in $[1, 3]$.

(d) Must every point of $[1, 3]$ be a limit point of such a sequence? Justify.

(e) True or false: (i) if $a_n \to L$, then every subsequence of (a_n) converges to L; (ii) every bounded sequence of reals has at least one limit point; (iii) if (a_n) is bounded, then $\limsup a_n$ is a limit point of (a_n).

Solution. (a) Let $\varepsilon > 0$. Choose N_1 with $|a_{2n} - L| \le \varepsilon$ for $n \ge N_1$ and N_2 with $|a_{2n-1} - L| \le \varepsilon$ for $n \ge N_2$. Put $N = 2\max\{N_1, N_2\}$. If $m \ge N$ is even, $m = 2n$ with $n \ge N_1$; if $m \ge N$ is odd, $m = 2n - 1$ with $n \ge N_2$. In both cases $|a_m - L| \le \varepsilon$.

(b) $a_n = (-1)^{n+1}$ has $a_{2n} = -1 \to -1$, but (a_n) alternates between 1 and -1 and does not converge. Both the even and odd subsequences must converge to the same limit.

(c) (i) 3 *is a limit point.* Fix $\varepsilon > 0$ and $N \ge 1$. Since $3 = \inf_M a_M^+$, the number $3 + \varepsilon$ is not a lower bound of $\{a_M^+\}$, so there is M_0 with $a_{M_0}^+ < 3 + \varepsilon$, whence $a_n < 3 + \varepsilon$ for all $n \ge M_0$. Note: M_0 depends only on ε, *not* on N. Also $a_P^+ \ge 3$ for every P (an infimum is below each member), so $3 - \varepsilon$ is not an upper bound of the tail $\{a_n : n \ge P\}$: for every P there is $n \ge P$ with $a_n > 3 - \varepsilon$. Take $P = \max\{M_0, N\}$: there is $n \ge P \ge N$ with $3 - \varepsilon < a_n < 3 + \varepsilon$, so $|a_n - 3| \le \varepsilon$. Hence 3 is a limit point. By the recursive construction (Problem 18 and the lecture: choose n_1 with $|a_{n_1} - 3| \le 1$ and $n_{j+1} > n_j$ with $|a_{n_{j+1}} - 3| \le \frac{1}{j+1}$) there is a subsequence converging to 3. The same argument with $\liminf$ gives a subsequence converging to 1.

★ **KEY METHOD** *(L18)*: **"An M_0 independent of N": combine an eventual bound with an infinitely-often bound using $P = \max\{M_0, N\}$.** This is the heart of the lecture's proof of Bolzano-Weierstrass. From $L + \varepsilon > \inf a_M^+$ you get an index M_0 *depending only on* ε such that all later terms are $< L + \varepsilon$. For each N you also have a term $> L - \varepsilon$ beyond *any* starting point. Taking the starting point $P = \max\{M_0, N\}$ gives a term with $n \geq N$ that satisfies both inequalities.

(ii) Let $a_{n_j} \to v$. Given $\varepsilon > 0$ there is M_0 with $a_n \leq 3 + \varepsilon$ for $n \geq M_0$ (as in (i)), and similarly $a_n \geq 1 - \varepsilon$ for all $n \geq M_0'$. Since $n_j \geq j$, the terms a_{n_j} eventually lie in $[1 - \varepsilon, 3 + \varepsilon]$, so $v \in [1 - \varepsilon, 3 + \varepsilon]$ for all $\varepsilon > 0$, i.e. $v \in [1, 3]$.

(d) No. Take $a_n = 1$ for odd n and $a_n = 3$ for even n: $\limsup = 3$, $\liminf = 1$, but the only limit points are 1 and 3, not (say) 2.

(e) (i) **True.** If (a_{n_j}) is a subsequence then $n_j \geq j$. Given $\varepsilon > 0$ and N with $|a_n - L| \leq \varepsilon$ for $n \geq N$, we get $|a_{n_j} - L| \leq \varepsilon$ for all $j \geq N$.

(ii) **True.** By Bolzano-Weierstrass (Problem 23) it has a convergent subsequence, and the limit of a convergent subsequence is a limit point.

(iii) **True.** Exactly the argument of (c)(i) with 3 replaced by $\limsup a_n$ shows that for every $\varepsilon > 0$ and N there is $n \geq N$ with $|a_n - \limsup a_n| \leq \varepsilon$.

★ **KEY METHOD** *(L18)*: **A strictly increasing $f : \mathbb{N} \to \mathbb{N}$ satisfies $f(n) \geq n$ (induction).** Base: $f(1) \geq 1$. Step: $f(n+1) > f(n) \geq n$ gives $f(n+1) \geq n + 1$. This one line is what makes "every subsequence of a convergent sequence converges to the same limit" work, and it is used in the lecture proof that limit points and subsequential limits coincide.

Problem 23. *Every sequence has a monotone subsequence* [L16, L18] ⋆⋆⋆

Call an index n a *peak* of a real sequence (a_n) if $a_n \geq a_m$ for every $m > n$.

(a) If there are infinitely many peaks, show (a_n) has a decreasing subsequence.

(b) If there are only finitely many peaks, show (a_n) has a strictly increasing subsequence.

(c) Deduce that every real sequence has a monotone subsequence.

(d) Deduce the Bolzano-Weierstrass theorem: every bounded real sequence has a convergent subsequence.

(e) Is the conclusion of (d) still true for an unbounded sequence? Give an example and a counterexample.

Solution. (a) List the peaks as $n_1 < n_2 < \cdots$. By definition of a peak, $a_{n_1} \geq a_{n_2}$ since $n_2 > n_1$, and $a_{n_2} \geq a_{n_3}$, and so on. So (a_{n_j}) is decreasing.

(b) Let n_1 be larger than every peak. Then n_1 is not a peak, so there is $n_2 > n_1$ with $a_{n_2} > a_{n_1}$. Since $n_2 > n_1$ is also not a peak, there is $n_3 > n_2$ with $a_{n_3} > a_{n_2}$, and so on. This builds a strictly increasing subsequence.

(c) Either there are infinitely many peaks or finitely many; the two cases are (a) and (b).

(d) If (a_n) is bounded, the monotone subsequence from (c) is bounded and monotone, hence convergent (bounded monotone sequences converge, Lecture 16).

(e) For an unbounded sequence, the conclusion can hold or fail: $a_n = n$ for odd n and $a_n = 0$ for even n has the convergent subsequence $(0, 0, \dots)$, while $a_n = n$ has no convergent sub-

sequence (every subsequence is unbounded). Hence boundedness is needed for the theorem, although (c) holds without it ($a_n = n$ itself is monotone).

★ **KEY PROOF TECHNIQUE**: **Case split on "infinitely many" versus "finitely many".** Whenever a construction depends on whether a set of indices is infinite, split on that dichotomy; "finitely many" lets you start beyond the last exceptional index (here: beyond all peaks), "infinitely many" lets you list them in order.

★ **KEY METHOD** *(L18)*: **Recursive construction of a subsequence.** Again: choose n_1, then given n_j find $n_{j+1} > n_j$ using the property valid beyond n_j. Two different monotone subsequences were produced in (a) and (b) by the same scheme.